

# Hybrid Time-Frequency Domain Equalization for Single-Carrier Underwater Acoustic Communications

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## ABSTRACT

Single-carrier block transmission (SCBT) has a great potential for providing higher bandwidth efficiency over frequency-selective channels with reduced computational complexity and low peak-to-average power ratio (PAPR). In this paper we propose a hybrid time-frequency domain equalization method for pseudorandom noise (PN) based single-carrier block transmissions. The proposed hybrid time-frequency domain equalizer includes two parts: a multi-channel frequency domain equalizer and a single channel adaptive time domain decision feedback equalizer (DFE). Bidirectional DFE and time-reversal diversity processing to cancel both residual post-cursor and pre-cursor intersymbol interference (ISI) is extended to hybrid time-frequency domain equalizer.

Underwater acoustic communication systems with single transmitter and multiple receivers have been designed and tested in lake in 2011. At a distance of 1.8 km, the useful data rates around 1500-7000 bps were achieved with uncoded bit error rate (BER)  $10^{-3}$ - $10^{-5}$  using the presented schemes. Experiment shows the performance can be enhanced by 1.5-3.8 dB in terms of output signal-to-noise ratio (SNR). A sea experiment was conducted with single transmitter and single receiver in 2012. At a distance of 625 m, the useful data rates around 8750 bps were achieved with uncoded bit error rates  $10^{-2}$ - $10^{-3}$  using the presented schemes. Experiment shows the performance can be enhanced by 1.6-2 dB in terms of output SNR.

## Categories and Subject Descriptors

H.4 [Information Systems Applications]: Communication Applications; C.3 [Special-Purpose and Application-based Systems]: Signal Processing Systems

## General Terms

Algorithms, Design, Experimentation, Measurements

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## Keywords

underwater acoustic communication, single-carrier block transmission, hybrid time-frequency domain equalization, bidirectional DFE, time-reversal diversity

## 1. INTRODUCTION

With the development of ocean exploring and environment monitoring, high-rate underwater acoustic communication (UAC) for underwater platform has received much attention recently. With the characteristics of low sound speed, high transmission loss, limited available bandwidth, complex multipath propagation and high background noise, underwater channel is one of the most difficult digital communication channels. For UAC, modulation method is the core technology that decides system performance, bandwidth efficiency and receiver complexity, etc. [1]. Over the last two decades, modulation methods and corresponding receiver algorithms play a key role in UAC, from  $M$ -ary frequency shift keying (MFSK) technology [1],  $M$ -ary phase shift keying (MPSK) [2, 3, 4, 5] to orthogonal frequency division multiplexing (OFDM) [6, 7, 8, 9] and spread spectrum technologies.

High-rate UAC based on coherent modulation is still a challenging problem. So far, high-rate UAC methods in the literature are mainly divided into two categories. The first one is based on single carrier adaptive multi-channel time-domain decision feedback equalizer (SC-TDE) with an embedded second order digital phase locked loop (DPLL) to simultaneously track carrier phase as well as mitigate multipath channel distortions proposed by Stojanovic[2]. SC-TDE receiver is sensitive to the equalizer parameters, such as the step size and parameters of the DPLL, and also quite demanding in terms of computational complexity. There are several methods for solving these problems, including time-reversal underwater acoustic communication [4], beam-domain multi-channel decision feedback equalizer [5], channel estimation based equalizer [3], correlation-based DFE [10] and various low complexity adaptive algorithms. The second one is based on OFDM. One key shortcoming of OFDM based UAC is the complexity of acoustic transducers (e.g., hydrophones) when OFDM signals exhibit high peak-to-average power ratio (PAPR). High PAPR will seriously affect the efficiency of power amplifier and transducer safety. Another drawback of OFDM is the difficulty in preserving orthogonal subcarriers under relative high Doppler shifts at acoustic wavelength.

Single-carrier block transmission technology (SCBT), through a single tap frequency equalizer, can obtain the same, or even better performance than OFDM, with close computa-

tional complexity [11, 12, 13]. In addition, SCBT signal has lower PAPR and is less sensitive to frequency offset than OFDM. Its calculation is approximately proportional to the logarithmic of the multipath extension length.

PN-based single carrier block transmission (PN-SCBT) has been proposed in our previous work [14]. It has some advantages over cyclic prefix SCBT (CP-SCBT) or zero padding SCBT (ZP-SCBT) using for shallow-water acoustic communications [15], in synchronization, channel estimation, Doppler shift estimation and multi-user detection.

To combat the residual ISI caused by channel variations or channel estimation error in each block, we investigate a hybrid time-frequency domain equalizer to combat the residual ISI. The proposed hybrid time-frequency domain equalizer is composed of two parts: a multi-channel frequency domain equalizer and a single channel adaptive time domain DFE. Bidirectional DFE and time-reversal diversity processing to cancel both residual post-cursor and pre-cursor ISI is extended to hybrid time-frequency domain equalizer. A lake and sea experiments were designed and conducted. Experiments show the effectiveness of the proposed methods.

## 2. SYSTEM MODEL

At the transmitter, the data are divided into serial blocks first, each having length  $N - P$ , and then PN code of length  $P$  is inserted into the end of each block. At the beginning of the data, additional PN code is inserted. The data packet structure is shown in Fig. 1, where PN code can be viewed as a CP by the fast fourier transform (FFT) device and is always the same. If  $P$  is larger than multipath spread, inter-block interference (IBI) can be ignored. The packet structure can be expressed as,

$$\mathbf{x} = [x(0), x(1), \dots, x(N - P - 1), q(0), q(1), \dots, q(P - 1)]^T, \quad (1)$$

where  $q(i)$  is PN code with length  $P$ ;  $(\cdot)^T$  represents matrix transpose.

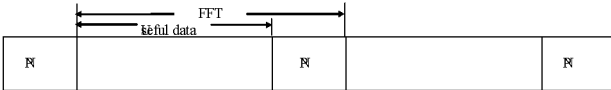


Figure 1: PN-SCBT packet structure

After synchronization at the receiver, the first PN signal is deleted. Then, the received signal is divided into blocks with length  $N$ . The data is converted back to frequency-domain from time-domain via FFT operation, and then frequency-domain equalization is conducted. The equalized signal is converted back to time-domain signal and time-domain signal, based on which decisions are made. The system model of PN-SCBT underwater acoustic communication is shown in Fig. 2.

For a time-invariant channel, the received signal can be expressed as

$$\mathbf{r} = \mathbf{h} \cdot \mathbf{x} + \mathbf{w}, \quad (2)$$

where

$$\mathbf{h} = \begin{bmatrix} h_0 & 0 & \dots & h_{L-1} & \dots & h_1 \\ h_1 & h_0 & \dots & \dots & \dots & h_2 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ h_{L-1} & \dots & h_1 & h_0 & 0 & 0 \\ 0 & \dots & \dots & \dots & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & h_{L-1} & \dots & h_1 & h_0 \end{bmatrix} \quad (3)$$

is an  $N \times N$  circulant matrix.  $\mathbf{r} = [r_0, r_1, \dots, r_{N-1}]^T$ ,  $\mathbf{w} = [w_0, w_1, \dots, w_{N-1}]^T$  is vector with length  $N$ .  $L$  is the channel order. After  $N$ -point FFT operation, the frequency-domain expression of the received signal is

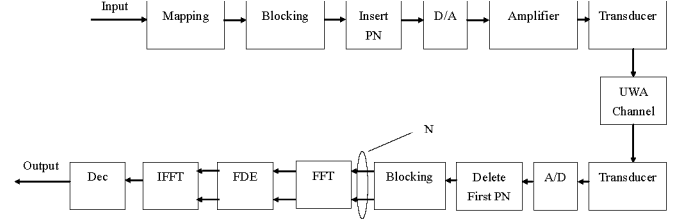


Figure 2: System model

$$\mathbf{R} = \Lambda \mathbf{X} + \mathbf{W}. \quad (4)$$

Since  $\mathbf{h}$  is a circulant Toeplitz matrix,  $\Lambda$  is a diagonal matrix. Its  $(k, k)$ -th entry as  $\Lambda_k$  is given as

$$\Lambda_k = \sum_{l=0}^{L-1} h(l) e^{-j(2\pi l k / N)}, \quad k = 0, \dots, N - 1. \quad (5)$$

With a single-tap equalizer on each frequency point, the frequency-domain equalized signal is

$$\hat{\mathbf{X}} = \mathbf{C} \mathbf{R}, \quad (6)$$

where  $\mathbf{C}$  is the equalization matrix.

## 3. RECEIVER ALGORITHMS

Suppose that the equivalent symbol spaced channel impulse response (CIR) is invariant over each block duration

$$h(k) = [h(k, 0), h(k, 1), \dots, h(k, L - 1)]^T, \quad (7)$$

where  $k$  denotes the  $k$ -th block,  $L$  is the channel order and  $T$  represents the matrix transpose. Due to the effect of the timing offset and Doppler shift, the received signal can be expressed as

$$r_k(n) = e^{j(2\pi f_k n T_s + \phi_{k,0})} \sum_{l=0}^{L-1} h(k, l) x_k(n - l) + w(k, n), \quad (8)$$

where  $T_s$  is data symbol duration;  $f_k$  is Doppler shift in  $k$ -th block, caused by relative motion between transceivers, A/D, D/A sampling, variant water current, etc.;  $\phi_{k,0}$  is phase rotation caused by symbol timing offset in  $k$ -th block;  $w(k, n)$  is additional white Gaussian noise with power  $\sigma^2$ . The  $k$ -th received block is expressed as

$$\mathbf{r}(k) = [r(k, 0), r(k, 1), \dots, r(k, N - 1)]^T. \quad (9)$$

For simplicity, without loss of generality, we eliminate the block index  $k$ . The received data block is

$$\mathbf{r} = \mathbf{D}\mathbf{h}\mathbf{x} + \mathbf{w}, \quad (10)$$

where  $\mathbf{w} = [w(0), w(1), \dots, w(N-1)]^T$  is noise vector, and  $\mathbf{D}$  is a diagonal rotated-phase matrix caused by Doppler shift, expressed as

$$\mathbf{D} = \text{diag} \left\{ e^{j\phi_{k,0}}, e^{j(2\pi f_k 2T_s + \phi_{k,0})}, \dots, e^{j(2\pi f_k 2(N-1)T_s + \phi_{k,0})} \right\}. \quad (11)$$

After the FFT, frequency-domain equalization and IFFT operation for  $\mathbf{r}$ , we get

$$\begin{aligned} \mathbf{z} &= \mathbf{F}^H \mathbf{C} \mathbf{F} \mathbf{r} \\ &= \mathbf{F}^H \mathbf{C} \Lambda \mathbf{F} \mathbf{D} \mathbf{x} + \mathbf{F}^H \mathbf{C} \mathbf{F} \mathbf{w} \\ &\approx \mathbf{I}_{N \times N} \mathbf{D} \mathbf{x} + \mathbf{F}^H \mathbf{C} \mathbf{F} \mathbf{w}, \end{aligned} \quad (12)$$

where  $\mathbf{F}\mathbf{F}^H = \mathbf{I}_{N \times N}$ ,  $\mathbf{F}$  and  $\mathbf{F}^H$  is the normalized DFT /IDFT matrix of size  $N \times N$ , and  $\mathbf{F}$  is

$$F_{l,k} = \frac{1}{\sqrt{N}} e^{-j(2\pi l k / N)}, \quad 0 \leq l, k \leq N-1. \quad (13)$$

where, matrix  $\Lambda = \mathbf{F}\mathbf{h}\mathbf{F}^H$  is a diagonal matrix formed with the values of the DFT of the composite CIR, given by eq. (5).

$\mathbf{C}$  is an  $N \times N$  diagonal matrix with its  $k$ -th diagonal element as the frequency domain equalizer taps  $C_k$ ,  $k = 0, \dots, N-1$ . When frequency domain minimum mean square error (MMSE) equalization is considered, the equalizer taps can be expressed as

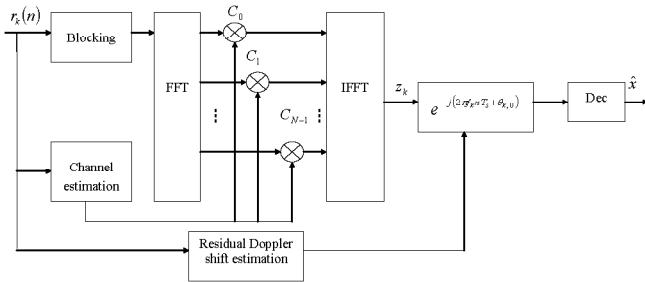
$$C_k = \frac{H_k^*}{|H_k|^2 + N_0/(2E_s)}, \quad (14)$$

where  $2E_s/N_0$  is the received signal-to-noise ratio (SNR).

After phase compensation for  $\mathbf{z}$  and decision, we get

$$\begin{aligned} \hat{\mathbf{x}} &= \text{dec}(\mathbf{D}^{-1}\mathbf{z}) \\ &= \text{dec}(\mathbf{x} + \mathbf{D}^{-1}\mathbf{F}^H \mathbf{C} \mathbf{F} \mathbf{w}). \end{aligned} \quad (15)$$

In summary, the receiver consists of synchronization, rotated-phase compensation, channel estimation, frequency domain equalization and decision (demapping). As shown in Fig. 3, this receiver is a block-by-block decoder.



**Figure 3: The multi-channel FDE receiver for PN-SCBT**

### 3.1 Synchronization

In PN-SCBT, PN signal is known at the receiver, and the length of PN signal is large enough in UAC to satisfy the synchronization accuracy. The start point of data block is given by the following equation using cross-correlation techniques:

$$\hat{\tau}_m = \arg \max_m \Re \left( \sum_{k=0}^{P-1} r(k+m) q^*(k) \right), \quad m = 1, 2, \dots, N_{ob}, \quad (16)$$

where,  $q(k)$  is the PN code of length  $P$ ,  $(\cdot)^*$  represents conjugate and  $N_{ob}$  is the observation length for synchronization.

### 3.2 Residual Doppler shift estimation

Two successive PN signals are used to estimate Doppler shift. In AWGN channel, the received PN signal of  $(k+1)$ -th block can be expressed as

$$\mathbf{R}_{k+1} = \mathbf{Q} e^{j\phi_{k+1,0}} + \mathbf{W}_{k+1}, \quad (17)$$

where,

$$\begin{aligned} \mathbf{R}_{k+1} &= \begin{bmatrix} r_{k+1}(0) \\ \vdots \\ r_{k+1}(P-1) \end{bmatrix}, \\ \mathbf{Q} &= \begin{bmatrix} q(0) \\ \vdots \\ q(P-1) \end{bmatrix}, \\ \mathbf{W}_{k+1} &= \begin{bmatrix} w_{k+1}(0) \\ \vdots \\ w_{k+1}(P-1) \end{bmatrix} \sim N(0, \sigma^2 \mathbf{I}_{P \times P}), \end{aligned} \quad (18)$$

and

$$\phi_{k+1,0} = \phi_{k,0} + N\Delta\phi_k. \quad (19)$$

Hence

$$\Delta\phi_k = \frac{\phi_{k+1,0} - \phi_{k,0}}{N}. \quad (20)$$

$\Delta\phi_k$  can be expressed as

$$\Delta\phi_k = \frac{1}{P} \angle \left( \sum_{n=0}^{P-1} r_k^*(n) r_{k+1}(n) \right). \quad (21)$$

Because the phase  $\Delta\phi$  is distinguished at  $[-\pi, \pi)$ , when the absolute phase rotation due to residual Doppler shifts, we have

$$|\Delta\phi| \leq \frac{\pi}{P}, \quad (22)$$

some modifications may be required to get  $\hat{\Delta\phi}_k$ . Now, the Doppler shift of  $k$ -th block can be expressed as

$$\hat{f}_{d,k} = \frac{\hat{\Delta\phi}_k}{2\pi N T_s}. \quad (23)$$

Then we can get the value of phase shift  $\phi_{k,n}$

$$\phi_{k,n} = \phi_{k,0} + n\hat{\Delta\phi}_k, \quad 0 \leq n \leq N-1. \quad (24)$$

After rotated-phase compensation, the  $k$ -th received signal can be expressed as

$$\hat{r}_k(n) = r_k(n) e^{-j\phi_{k,n}}, \quad 0 \leq n \leq N-1. \quad (25)$$

### 3.3 Channel estimation

A PN code, like the  $m$  sequence, has good characteristics of correlation. The chips can be expressed as  $q_n$ ,  $n = 0, \dots, P-1$ , with value equal  $\{+1, -1\}$ . The last  $P-L+1$  received PN chips are given as

$$\begin{bmatrix} r_L \\ r_{L+1} \\ \vdots \\ r_P \end{bmatrix} = \begin{bmatrix} q_L & q_{L-1} & \dots & q_2 & q_1 \\ q_{L+1} & q_L & \dots & q_3 & q_2 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ q_P & q_{P-1} & \dots & q_{L+1} & q_L \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_L \end{bmatrix} + \begin{bmatrix} w_L \\ w_{L+1} \\ \vdots \\ w_P \end{bmatrix} \quad (26)$$

where  $L$  is channel order,

$$\mathbf{r} = \mathbf{C}_p \mathbf{h} + \mathbf{w}, \quad (27)$$

and

$$\mathbf{C}_p = \begin{bmatrix} q_L & q_{L-1} & \dots & q_2 & q_1 \\ q_{L+1} & q_L & \dots & q_3 & q_2 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ q_P & q_{P-1} & \dots & q_{L+1} & q_L \end{bmatrix}. \quad (28)$$

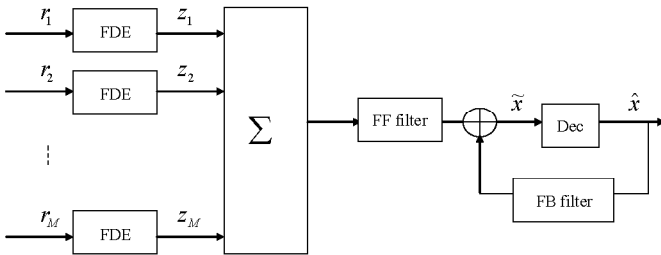
The least square (LS) channel estimation is

$$\mathbf{h}_{LS} = (\mathbf{C}_p^H \mathbf{C}_p)^{-1} \mathbf{C}_p^H \mathbf{r}. \quad (29)$$

### 3.4 Hybrid Time-Frequency Domain Equalizer

The basic FDE assumes the channel is invariant in one block. For channel variations in one block or channel estimation error, there is ISI in the output of FDE. To get spatial diversity, multi-channel frequency domain equalizers are used following a single channel DFE in order to combat the residual ISI.

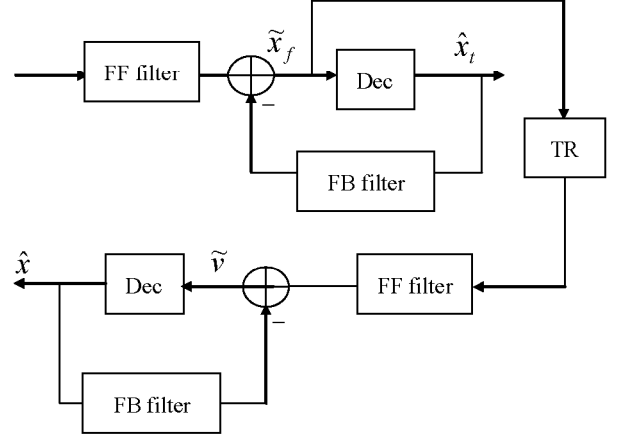
The DFE suffers from error propagation caused by the feedback of incorrect decisions. In order to reduce error propagation and improve the performance of hybrid time-frequency domain equalizer, bidirectional decision feedback equalizer (BiDFE) was first introduced in wireless channels [17], and in CCK-based underwater acoustic channels as [16] and later in time-reversal underwater acoustic communications [18].



**Figure 4: Hybrid time-frequency domain equalizer structure**

The BiDFE consists of two serial DFE structures, a forward DFE and a backward DFE. The forward one equalizes

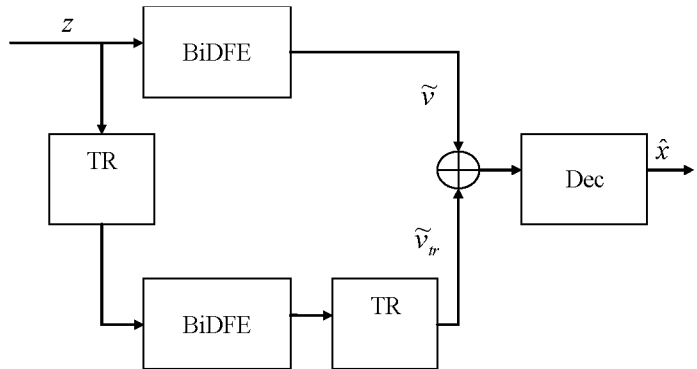
the received signal to cancel post-cursor ISI effect and make tentative decisions. On the other hand, to cancel precursor ISI caused by future symbols, we can perform a packet time-reversal (TR) processing. A TR is achieved by reversing the sequential order of the received samples in time. As a result, the equivalent CIR as seen by the equalizer becomes a TR of the actual CIR. In this way, the postcursor ISI in the original sequence becomes pre-cursor ISI in the TR sequence, while the pre-cursor ISI in the original sequence becomes post-cursor ISI in the TR sequence. Once again, we can cancel these ISIs in the TR sequence by applying a backward DFE.



**Figure 5: Bidirectional DFE**

The BiDFE assumes the condition of correct past symbol feedback. In practice, however, erroneous decisions can happen. As a result, the pre-cursor and post-cursor ISI cancellations can in fact enhance the ISI effect. This can induce an error propagation effect which causes performance degradation.

To mitigate the effect due to errors in the tentative decisions, iterative processing and TR diversity can be useful for performance improvement. The estimated symbols are regarded as tentative decision for both post-cursor and pre-cursor ISI cancellations.



**Figure 6: Time-reversal diversity**

The basic assumption of TR diversity is that error propagation is causal, and therefore, the errors caused by the forward DFE and the backward DFE have low correlation. This low-correlation characteristic leads to inherent diversity gain. In this paper, we exploit the TR-diversity in our BiDFE to achieve additional diversity gain, thereby improving the receiver BER performance. To generalize, we include a forward BiDFE and a backward BiDFE in iterative TR diversity receiver as shown in Fig. 6. We adopt equal-gain combining strategy for the forward BiDFE soft output  $v(k)$  and the backward BiDFE output  $v_{tr}(k)$ .

## 4. EXPERIMENTAL RESULTS

### 4.1 Lake experiment

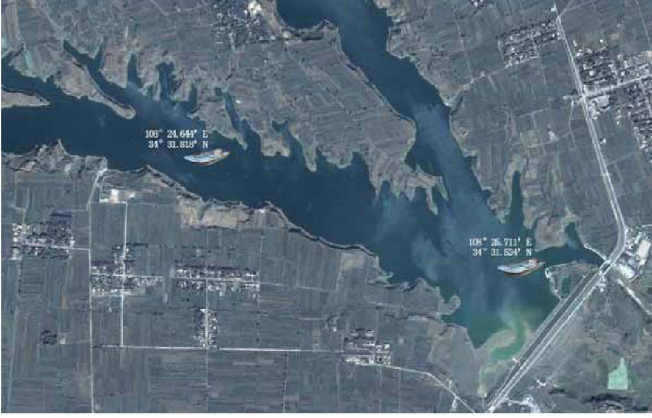


Figure 7: Lake experiment site. The distance between transmitter and receiver is about 1.8 km

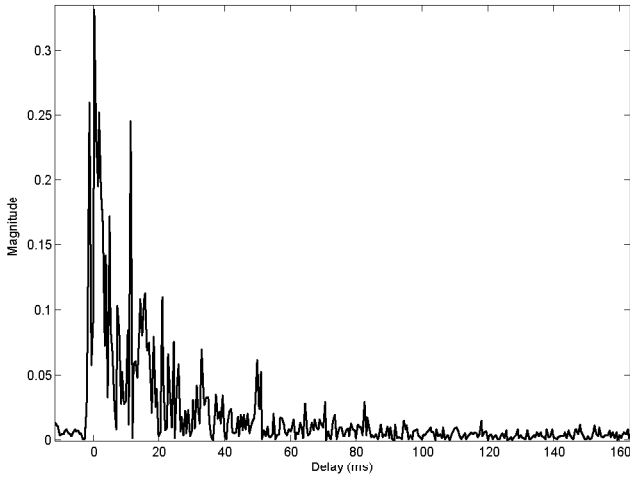


Figure 8: Lake experiment channel structure

In order to verify the performance of the proposed PN-SCBT method in practical channel, experiment was conducted in a practical lake of Shaanxi Province in 2011. The water is with a depth of 4 m and 12 m at the transmitter and receiver sites respectively. The transmitter transducer

was deployed below the surface of 1 m, and the receiver with six hydrophones was deployed below the surface of 3 m. The distance between transmitter and receiver was 1.8 km.

The transmitted signal is structured in frames. Each frame consists of 60 short blocks. In each block, a PN of length 128 is inserted to the transmitted signal and form a whole block as shown in Fig. 2. Each frame is headed by a linear frequency modulation (LFM) signal for synchronization. The symbol duration is 0.5 ms, with QPSK, 8PSK and 16QAM modulation for  $N = 1024$  and  $P = 128$ . The carrier frequency is 9 kHz and the signal bandwidth is 2 kHz. No pulse shaping filter is used in this test. The corresponding useful data rates are around 3500, 5250 and 7000 bit/s. Fractionally spaced equalizers (T/2) are used for frequency domain equalizer. The recursive least squares (RLS) algorithm is used for the adaptive single channel DFE.

The estimated channel using LFM is shown in Fig.8. There are several strong multipath interferences around the strongest path. The dense multipath spread is about 50 ms (100 symbols).

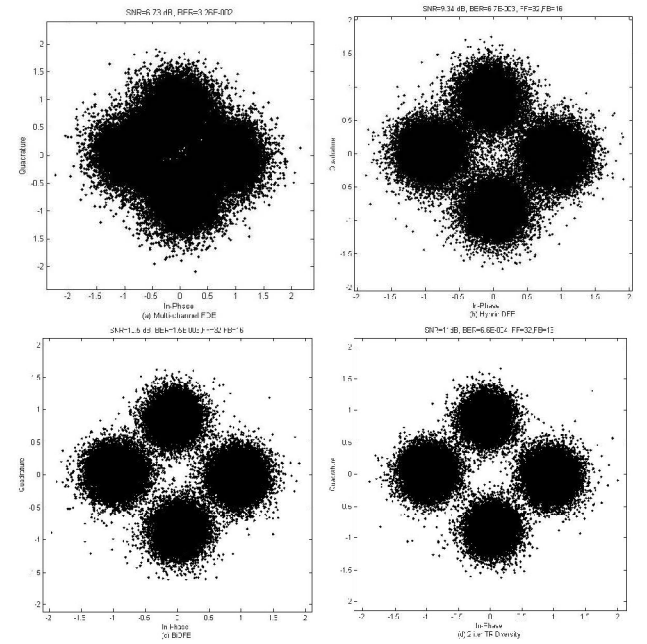


Figure 9: Performance of different equalizers for SCBT with QPSK modulation

Fig. 9 shows the results of above schemes for SCBT with QPSK modulation, from top to bottom and left to right: (a) normal multi-channels frequency domain equalizer, (b) hybrid time-frequency domain equalizer, (c) bidirectional DFE and (d) 2 iter TR Diversity. The data rate is 3500 bits/s with QPSK modulation. The numbers of feedforward filter and feedback filter taps in both direction are 32 and 16 respectively. The output SNRs are 6.73 dB, 9.34 dB, 10.5 dB, 11 dB and the BERs are  $3.26 \times 10^{-2}$ ,  $6.7 \times 10^{-3}$ ,  $1.5 \times 10^{-3}$  and  $6.6 \times 10^{-4}$ , respectively. Hybrid time-frequency equalization with BiDFE and TR can improve the normal multi-channel frequency domain equalization by 2.6-4.3 dB in terms of output SNR for QPSK modulation.

Fig. 10 shows the results of above schemes for SCBT with

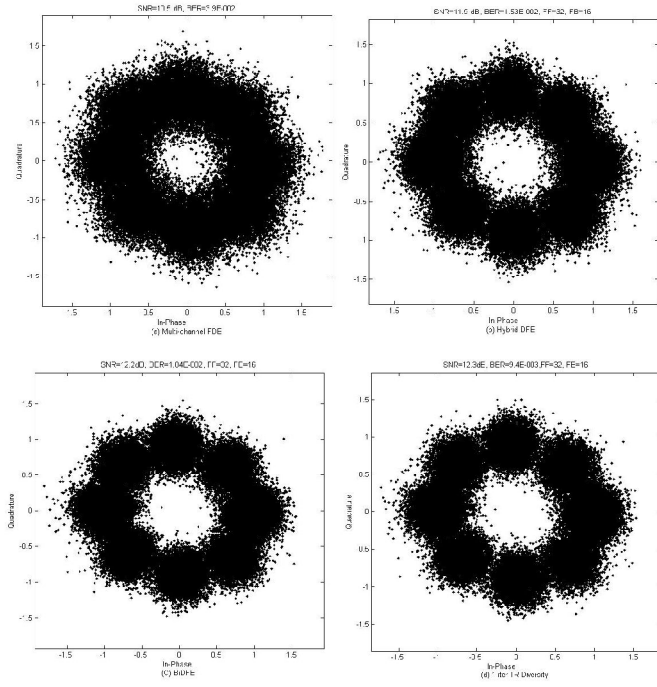


Figure 10: Performance of different equalizers for SCBT with 8PSK modulation

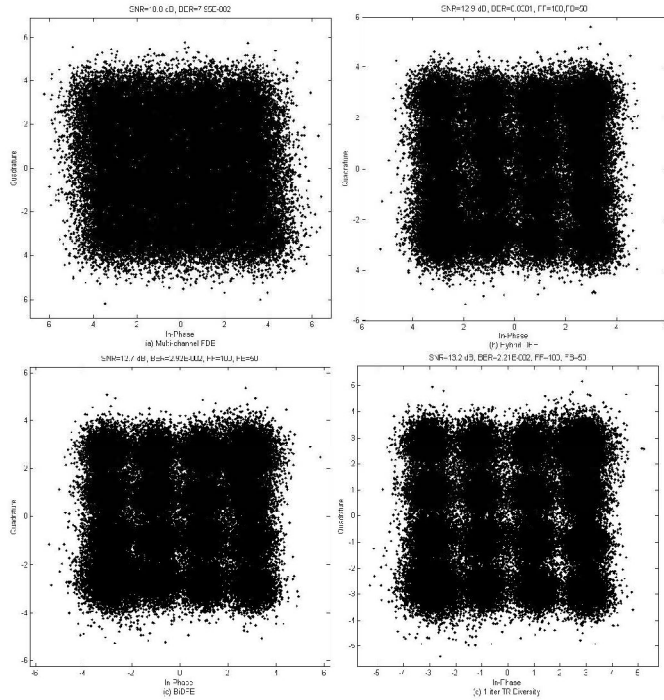


Figure 11: Performance of different equalizers for SCBT with 16QAM modulation

8PSK modulation, from top to bottom and left to right: (a) normal multi-channels frequency domain equalizer, (b) hybrid time-frequency domain equalizer, (c) bidirectional DFE and (d) 2 iter TR Diversity. The data rate is 5250 bits/s. The numbers of feedforward filter and feedback filter taps in both directions are 32 and 16 respectively. The output SNRs are 10.5 dB, 11.9 dB, 12.3 dB, 12.3 dB and the BERs are  $3.9 \times 10^{-2}$ ,  $1.53 \times 10^{-2}$ ,  $1.04 \times 10^{-2}$  and  $9.4 \times 10^{-3}$ , respectively. Hybrid time-frequency equalizations with BiDFE and TR can improve the normal multi-channel frequency domain equalization by 1.4-1.8 dB in terms of output SNR for 8PSK modulation.

Fig. 11 shows the results of above schemes for SCBT with 16QAM modulation, from top to bottom and left to right: (a) normal multi-channels frequency domain equalizer, (b) hybrid time-frequency domain equalizer, (c) bidirectional DFE and (d) 2 iter TR Diversity. The data rate is 7000 bits/s. The numbers of feedforward filter and feedback filter taps in both directions are 100 and 50, respectively. The output SNRs are 10.8 dB, 12.9 dB, 12.7 dB, 13.2 dB and the BERs are  $7.95 \times 10^{-2}$ ,  $3.81 \times 10^{-2}$ ,  $2.92 \times 10^{-2}$  and  $2.21 \times 10^{-2}$ . Hybrid time-frequency equalizations with BiDFE and TR can improve the normal multi-channel frequency domain equalization by 2.1-2.4 dB in terms of output SNR for 16QAM modulation.

## 4.2 Sea experiment



Figure 12: Sea experiment site. The distance between transmitter and receiver is about 625 m

In order to verify the performance of the proposed PN-SCBT method in practical channels, the sea experiment was conducted near Xiamen university on March 12, 2012 as shown in Fig. 12. The distance between transmitter and receiver was around 625 m. Only one hydrophone was used in the receiver.

The transmitted signal is structured in frames. Each frame consists of 68 short blocks. In each block, a PN of length



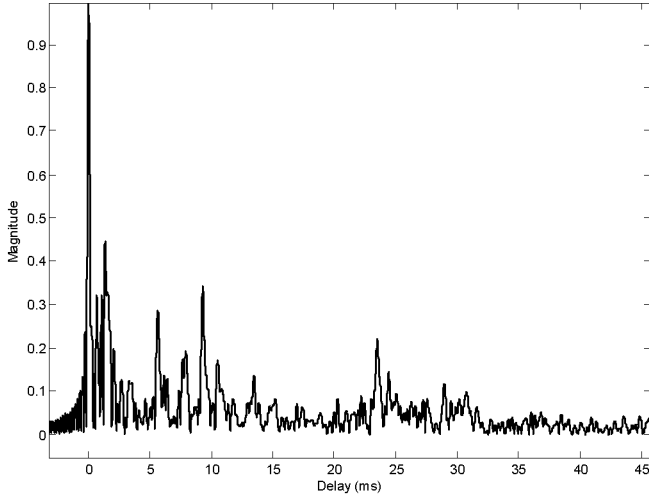


Figure 13: Sea experiment channel structure

512 is inserted to the transmitted signal and form a whole block as shown in Fig. 2. Each frame is headed by a linear frequency modulation (LFM) signal for synchronization. The symbol duration is 0.2 ms, with QPSK modulation for  $N = 4096$  and  $P = 512$ . The useful data is around 8750 bit/s, with bandwidth 10-20 kHz. The carrier frequency is 15 kHz. The pulse shaping filter is the root raised cosine filter with a roll-off factor of 0.25. Fractionally spaced equalizers ( $T/2$ ) are used for frequency domain equalizer. The RLS algorithm is used for the adaptive DFE.

The estimated channel using LFM is shown in Fig.13. The multipath spread is a sparse channel with about 25 ms (125 symbols) delay spread.

Fig. 14 shows the results of above schemes for SCBT with QPSK modulation at sea experiment from top to bottom and left to right: (a) normal multi-channels frequency domain equalizer, (b) hybrid time-frequency domain equalizer, (c) bidirectional DFE and (d) 2 iter TR Diversity. The data rate is 8750 bits/s with QPSK modulation. The numbers of feedforward filter and feedback filter taps in both direction are 100 and 50 respectively. The output SNRs are 6.3 dB, 7.9 dB, 8.2 dB, 8.3 dB and the BERs are  $3.43 \times 10^{-2}$ ,  $1.41 \times 10^{-2}$ ,  $9.2 \times 10^{-3}$  and  $6.6 \times 10^{-3}$  respectively. Hybrid time-frequency equalizations with BiDFE and TR can improve the normal multi-channel frequency domain equalization by 1.6-2 dB in terms of output SNR.

Table 1 shows the output SNR and BER improvements of different equalizers. For different modulations, the first line shows the output SNRs and the second line shows the BERs. The second column in the table is the performance of normal multi-channel FDE.

## 5. CONCLUSION

A PN-SCBT method over underwater acoustic channel and corresponding receiver algorithms were proposed in this paper. The lake experiment shows a large improvement in SNR and BER. For QPSK, the improvement from hybrid time-frequency domain equalizer to iterative TR diversity equalizer in SNR is larger than 4 dB. For 8PSK and 16QAM, the improvement from hybrid time-frequency domain equal-

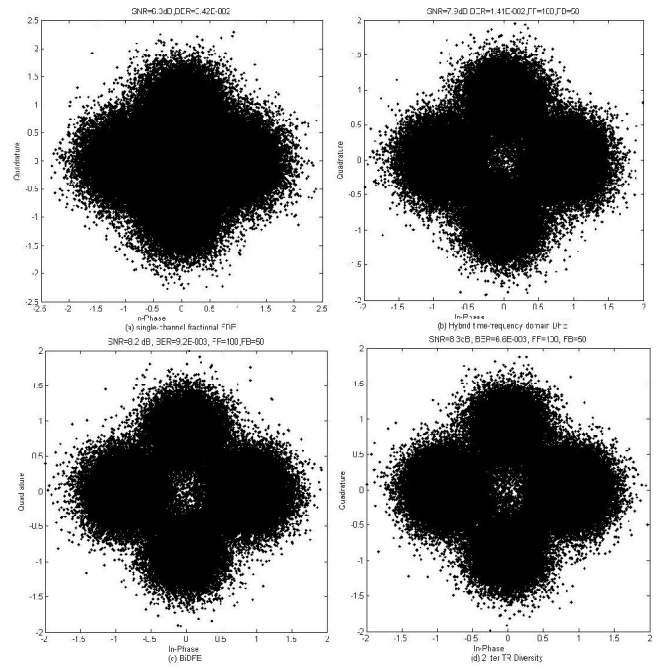


Figure 14: Performance of Hybrid Time-Frequency Equalizer for SCBT with QPSK modulation at sea experiment

izer to iterative TR diversity equalizer in SNR is within 0.4 dB. The sea experiment shows that hybrid time-frequency domain equalizer can get 1.6 dB improvement compare to the normal multi-channel FDE.

Table 1: SNR and BER performance of different equalizers

| Mod.             | FDE                   | Hybrid FDE            | BiDFE                 | TR-Diver.             |
|------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| QPSK             | 6.73                  | 9.34                  | 10.5                  | 11                    |
|                  | $3.26 \times 10^{-2}$ | $6.7 \times 10^{-3}$  | $1.5 \times 10^{-3}$  | $6.6 \times 10^{-4}$  |
| 8PSK             | 10.5                  | 11.9                  | 12.2                  | 12.3                  |
|                  | $3.9 \times 10^{-2}$  | $1.53 \times 10^{-3}$ | $1.04 \times 10^{-2}$ | $9.4 \times 10^{-3}$  |
| 16QAM            | 10.8                  | 12.9                  | 12.7                  | 13.2                  |
|                  | $7.95 \times 10^{-2}$ | $3.81 \times 10^{-3}$ | $2.92 \times 10^{-2}$ | $2.21 \times 10^{-2}$ |
| QPSK<br>Sea test | 6.3                   | 7.9                   | 8.2                   | 8.3                   |
|                  | $3.42 \times 10^{-2}$ | $1.41 \times 10^{-2}$ | $9.2 \times 10^{-3}$  | $6.6 \times 10^{-3}$  |

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